INF266 - Problem Set 4

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Part I - Problems

Exercise 1

1.

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma)$$

States

$$\mathcal{S} = \{s_0, s_1, s_2, s_3, s_4, s_5, s_6, s_7\}$$

$$s_0 : \text{empty}, \quad s_1 : \text{full}, \quad s_2 : \text{sugared}, \quad s_3 : \text{salted},$$
$$s_4 : \text{sugar sampling}, \quad s_5 : \text{salt sampling},$$
$$s_6 : \text{baked sugary}, \quad s_7 : \text{baked salty}$$

Actions

$$\begin{aligned}
&\mathcal{A}(s_0) = \{\text{wait}, \text{prep}\},\\
&\mathcal{A}(s_1) = \{\text{wait}, \text{sugar}, \text{salt}\},\\
&\mathcal{A}(s_2) = \mathcal{A}(s_3) = \{\text{taste}, \text{cook}, \text{restart}\},\\
&\mathcal{A}(s_4) = \mathcal{A}(s_5) = \{\text{cook}, \text{return}\},\\
&\mathcal{A}(s_6) = \mathcal{A}(s_7) = \emptyset
\end{aligned}$$

Transitions (deterministic)

$$P(s'|s,a) = 1 \text{ for:}$$

$$\begin{aligned}
&(s_0, \text{wait}) \to s_0,\ (s_0, \text{prep}) \to s_1\\
&(s_1, \text{sugar}) \to s_2,\ (s_1, \text{salt}) \to s_3\\
&(s_2, \text{taste}) \to s_4,\ (s_2, \text{cook}) \to s_6,\ (s_2, \text{restart}) \to s_1\\
&(s_3, \text{taste}) \to s_5,\ (s_3, \text{cook}) \to s_7,\ (s_3, \text{restart}) \to s_1\\
&(s_4, \text{cook}) \to s_6,\ (s_4, \text{return}) \to s_2\\
&(s_5, \text{cook}) \to s_7,\ (s_5, \text{return}) \to s_3
\end{aligned}$$

Reward

$$R(s,a,s') = \begin{cases} +10 & s' = s_6 \\ -10 & s' = s_7 \\ 0 & \text{otherwise} \end{cases}$$

$$\gamma \in [0,1]$$

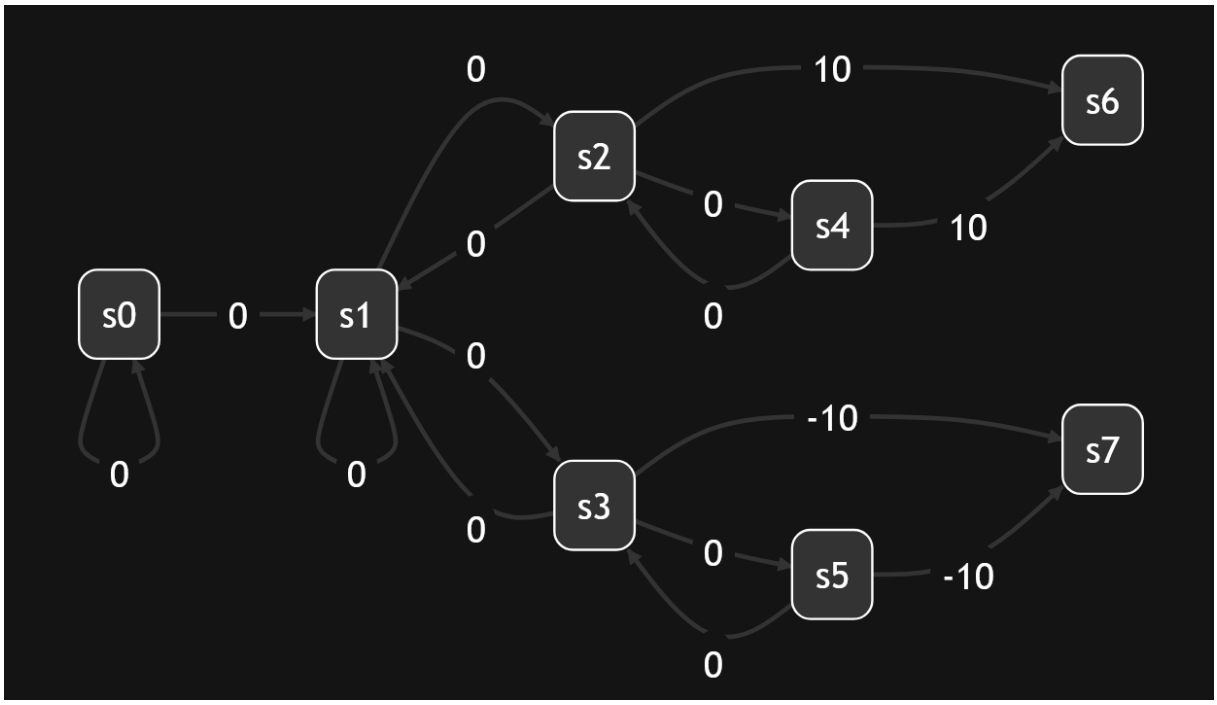

Figure 1: Backup diagram for the MDP.

2.

3. **y**es, we can define an optimal deterministic policy for this MDP. The optimal policy would be:

   (a) In state s_0, take action prep to move to state s_1.

   (b) In state s_1, take action sugar to move to state s_2.

   (c) In state s_2, take action cook to move to state s_6 and receive a reward of $+10$.

   This policy is optimal because it leads to the highest possible reward of $+10$ by reaching state s_6. Any deviation from this policy would either lead to a lower reward (e.g., going to state s_7 with a reward of -10), a delayed reward (like taking action taste in s_2 leading to state s_4), or no reward at all (e.g., waiting in state s_0 or taking the restart action in states s_2 or s_3). Therefore, the optimal deterministic policy is to always choose the actions that lead to state s_6. This policy is simply the shortest path to the highest reward.

4. **Y**es, an optimal stochastic policy exists. However, since the environment is fully deterministic and there is a unique optimal trajectory leading to reward $+10$, any randomness that assigns non-zero probability to suboptimal actions (e.g., adding salt) will reduce the expected return. Thus, the optimal stochastic policy is actually just deterministic, meaning the agent always chooses the best action with probability 1 and never chooses any other actions.

$$\pi(a \mid s) = \begin{cases} 1 & \text{if } a = a^*(s), \\ 0 & \text{otherwise.} \end{cases}$$

   In particular, the agent always chooses actions that lead to the sugary cookies and avoids all actions leading to the salty outcome.

5. We define the MDP as

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma).$$

   States To retain the Markov property, we include the latent variable $\sigma \in \{\text{sugar}, \text{salt}\}$:

$$\mathcal{S} = \{s_0, s_1, (s_2, \sigma), s_4, s_5, s_6, s_7\},$$

where

$$
\begin{aligned}
s_0 &: \text{empty bowl}, \quad s_1 : \text{full bowl}, \\
(s_2, \sigma) &: \text{enriched bowl with ingredient } \sigma, \\
s_4 &: \text{sugar sampling}, \quad s_5 : \text{salt sampling}, \\
s_6 &: \text{baked sugary cookies}, \quad s_7 : \text{baked salty cookies}.
\end{aligned}
$$

Actions

$$
\begin{aligned}
\mathcal{A}(s_0) &= \{\text{wait}, \text{prepare dough}\}, \\
\mathcal{A}(s_1) &= \{\text{wait}, \text{add ingredient}\}, \\
\mathcal{A}((s_2, \sigma)) &= \{\text{taste}, \text{cook}, \text{restart}\}, \\
\mathcal{A}(s_4) = \mathcal{A}(s_5) &= \{\text{cook}, \text{return}\}, \\
\mathcal{A}(s_6) = \mathcal{A}(s_7) &= \emptyset.
\end{aligned}
$$

Transitions

$$
\begin{aligned}
&(s_0, \text{wait}) \to s_0, \quad (s_0, \text{prepare dough}) \to s_1, \\
&(s_1, \text{add ingredient}) \to \begin{cases} (s_2, \text{sugar}) & \text{with prob } 0.5, \\ (s_2, \text{salt}) & \text{with prob } 0.5, \end{cases} \\
&((s_2, \sigma), \text{taste}) \to \begin{cases} s_4 & \text{if } \sigma = \text{sugar}, \\ s_5 & \text{if } \sigma = \text{salt}, \end{cases} \\
&((s_2, \sigma), \text{cook}) \to \begin{cases} s_6 & \text{if } \sigma = \text{sugar}, \\ s_7 & \text{if } \sigma = \text{salt}, \end{cases} \\
&((s_2, \sigma), \text{restart}) \to \begin{cases} (s_2, \text{sugar}) & \text{with prob } 0.5, \\ (s_2, \text{salt}) & \text{with prob } 0.5, \end{cases} \\
&(s_4, \text{cook}) \to s_6, \quad (s_4, \text{return}) \to (s_2, \text{sugar}), \\
&(s_5, \text{cook}) \to s_7, \quad (s_5, \text{return}) \to (s_2, \text{salt}).
\end{aligned}
$$

Reward function

$$
R(s, a, s') = \begin{cases} +10 & \text{if } s' = s_6, \\ -10 & \text{if } s' = s_7, \\ 0 & \text{otherwise}. \end{cases}
$$

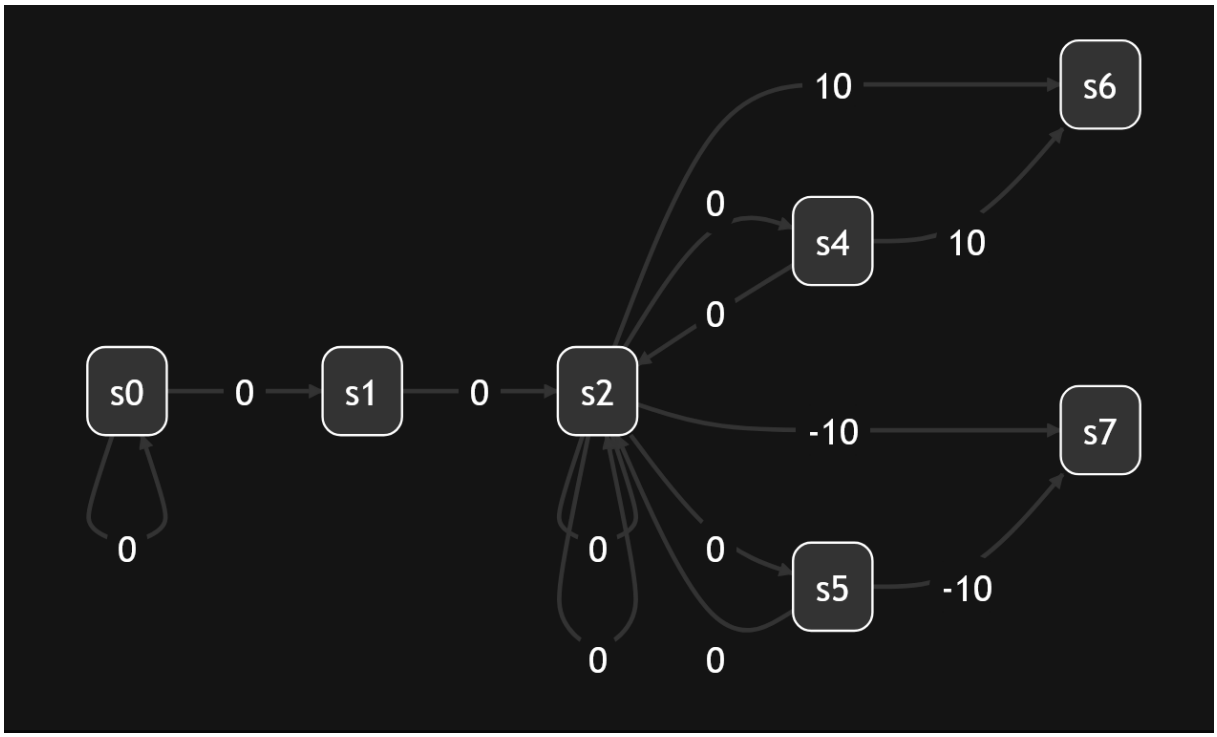

Figure 2: Backup diagram for the new MDP.

6.

7. No, an optimal deterministic policy does not exist. The agent cannot observe the latent variable $\sigma \in \{\text{sugar}, \text{salt}\}$, so in the enriched bowl state it cannot distinguish between the two underlying cases. A deterministic policy must choose the same action regardless of σ. However, the optimal behavior depends on information: the agent should first *taste* to reveal σ, and then choose *cook* if $\sigma = \text{sugar}$ or *restart* if $\sigma = \text{salt}$. This requires conditioning actions on information that is not available in the current observation. Thus, no deterministic memoryless policy can achieve optimal performance. An optimal policy must either use memory (history-dependent) or be stochastic.

8. Yes, an optimal stochastic policy exists.

   Due to partial observability, the agent must act under uncertainty in the enriched bowl state. The optimal strategy is to first gather information and then act optimally based on that information.

   An optimal policy is:

   $$\pi(\text{taste} \mid \text{enriched bowl}) = 1,$$

   and then

   $$\pi(\text{cook} \mid \text{sugar sampling}) = 1, \quad \pi(\text{return} \mid \text{salt sampling}) = 1.$$

   After returning to the enriched bowl from salt sampling, the agent again follows:

   $$\pi(\text{taste} \mid \text{enriched bowl}) = 1.$$

   Thus, the agent repeatedly tastes until it identifies sugar, and only then chooses to cook, ensuring the maximal expected reward.

9. The key difference between the original MDP and the new MDP is that the new MDP introduces partial observability through the latent variable σ that determines whether the ingredient added is sugar or salt. In the original MDP, the agent has full observability and can directly choose actions based on the current state. In contrast, in the new MDP, the agent cannot directly observe σ and must rely on actions (like tasting) to gain information about it. This means that an optimal policy may require memory or stochasticity, as the agent needs to condition its actions based on the information gathered.

10. Yes, a natural modification to the original MDP is to make the latent variable σ observable, but only after taking a specific action (e.g., tasting). This creates a scenario where the agent must first take an exploratory action to gain information about the state of the environment before it can make an informed decision. This would restore full observability while still requiring the agent to gather information. Alternativly we can augment the state space to incude the history of actions take, allowing the agent to infer the latent variable from past actions and observations. In both cases we remove partial observability and simplify the policy design.

Exercise 2